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# WM-Concord: Do World Model Diagnostics Agree?

A Cross Diagnostic Validity Study of World Model Diagnostics

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**Abstract**

*Four families of diagnostics are used to claim that a sequence model has, or lacks, a "world model": next token validity, state equivalence based on the Myhill Nerode theorem, inductive bias probes, and representation probing. We find that on Othello* ***they agree mainly because they all track how competent a model is, none of them reliably predicts when a model will break, and the one link to fragility comes from the training data****. WM-Concord runs all four on one controlled zoo of 81 small sequence models (transformer, LSTM and Mamba; three scales; three training distributions; three seeds) with a known ground truth, under a preregistered analysis. Raw agreement is strong (Kendall W 0.87), but with the validation legal move rate controlled it falls to W 0.27, and no pair among the three diagnostics that can be separated from competence still agrees. The remaining disagreement is not explained by architecture, scale or training data. Only the representation probe adds to the legal move rate in predicting fragility on shifted tasks, and an exploratory check traces this to the models trained on human championship games, which are both the most fragile and the least probe readable; among comparable models the link disappears. A paper that reports one diagnostic on models of different quality is therefore mostly reporting how good the models are. Navigation and a held out program interpreter follow with the same pipeline.*

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# 1. Introduction

Whether a sequence model trained only to predict the next token builds a coherent internal model of its domain is one of the central open questions in current machine learning. Several research groups have built diagnostics to answer it, but each group uses its own instrument, on its own models, in its own domain. The same Othello models have been reported both as having a world model [1, 2] and as lacking a coherent one [3, 4]. This project does not propose a new diagnostic. It measures whether the existing ones agree, and which one can be trusted.

**Contributions.**

- The first head to head comparison of four world model diagnostics on the same models and the same ground truth.
- A test of which diagnostic predicts failure on related but shifted tasks, judged against a fragility suite that is frozen before any diagnostic is run.
- An open source benchmark (WM-Concord) with one common interface, so a future diagnostic can be compared against these four on identical ground truth.

**What we found so far.** The headline is a confound: **the four diagnostics agree because they all track competence, and none of them reliably predicts fragility; the one link to fragility we find comes from the training data.** On the first domain, Othello, we trained 81 models (three architectures, three scales, three training distributions, three seeds), all of which passed a preregistered admission rule, and scored every one with all four diagnostics and two versions of a fragility suite. The diagnostics agree on which models are better, but they agree because all of them follow competence, the model's legal move rate: once competence is controlled for, the agreement largely disappears (Section 7.3). Where they disagree, the disagreement is consistent with seed noise (Section 7.4). No diagnostic is a reliable predictor of fragility: the representation probe adds to the legal move rate under the preregistered test, but an exploratory analysis traces this to the models trained on human games, which are both the most fragile and the least probe readable (Section 7.5). The practical message for anyone using these tools is to report several diagnostics, always with a competence control, rather than trust any single one.

# 2. Problem Statement

The four diagnostic families studied in this project are:

**D1, next token validity.** The fraction of the model's proposed next tokens that are legal in the true state.

**D2, state equivalence.** Compression error (two histories that reach the same state are treated differently) and distinction error (two different states are treated the same), following [3].

**D3, inductive bias probes.** How the model adapts to small synthetic tasks generated from the true world model [4].

**D4, representation probing.** Whether the true state can be read from the model's internal activations by a probe, checked against a control task and by causal intervention [1].

We test four hypotheses:

**H1, concordance.** Do the four diagnostics rank the same models in the same order?

**H2, disagreement.** When they disagree, is the disagreement systematic (tied to architecture, scale, or training data) or just noise?

**H3, predictive validity.** Which diagnostic best predicts failure on a frozen suite of related but shifted tasks?

**H4, generalization.** Do the answers hold on a held out domain?

**Challenges.** The diagnostics measure different things (behaviour, adaptation, and internal activations), so their scores are not on a common scale; we therefore compare only rankings. The state equivalence metrics cannot be computed exactly in large domains and must be estimated by sampling. Probes can report structure the model does not actually use. And every diagnostic is likely to reward a better trained model, so agreement can come from competence alone. The design handles each of these in Section 5.

# 3. Related Work

## 3.1. Representation probing

Li et al. [1] trained a GPT model on Othello move sequences only. The model never sees a board or the rules, yet it proposes legal moves almost every time, and probes trained on its internal activations recover the board state. They then edited the decoded board inside the model and showed that its next move predictions changed to match, which is evidence that the representation is used and not only present. Yuan and Søgaard [2] extended this to seven language models fine tuned on Othello and found similar board representations across all of them, which they read as support for a real world model. Both works use probing as the only instrument. WM-Concord uses probing as one of four diagnostics, adds a control task and a causal check, and compares its verdict with the other three on the same models.

Nanda et al. [5] returned to the same Othello model and showed that the board is in fact linearly represented, but relative to the player whose turn it is ("mine" and "theirs") rather than in absolute colours. With these labels a single linear layer reads the board almost perfectly, and adding or subtracting a probe direction inside the model flips its view of a square and changes its next move predictions accordingly. Our reproduction gate uses this result: our linear probes match their published accuracies within one percentage point on five of six values. WM-Concord takes their mine and theirs labels as the default for diagnostic D4, and asks whether a model that passes this probe also passes the three diagnostics that do not look inside the model.

Toshniwal et al. [6] used chess as a testbed for state tracking in language models. They trained transformers on games written as move sequences and tested state tracking with prompts that ask for the starting or ending square of a piece at a given point in the game, so the board is read from the model's own predictions rather than from its activations. They found that state tracking improves with more training data and when some training positions include the type of the piece that moved. Their evaluation is a behavioural one, close in spirit to next token validity. WM-Concord keeps such behavioural tests (D1) and places them next to internal (D4) and structural (D2, D3) tests on the same models.

## 3.2. State equivalence metrics

Vafa et al. [3] define the domain as a deterministic finite automaton and derive two metrics from the Myhill Nerode theorem. Compression error measures whether the model gives the same continuations to two histories that end in the same state; distinction error measures whether it can tell apart histories that end in different states. On taxi routes in Manhattan, Othello, and logic puzzles, models that looked good on next token accuracy failed these metrics, and a map reconstructed from the navigation model contained streets that do not exist. Models trained on

random walks built more coherent world models than models trained on shortest paths. That work compares its metrics only with next token accuracy. WM-Concord uses them as diagnostic D2, adopts their three training distributions, and compares them with all three other diagnostics and with an external fragility criterion.

### 3.3. Inductive bias probes

Vafa et al. [4] ask a different question: does a model's inductive bias point at the true world model? They adapt a pretrained model to small synthetic datasets generated from a postulated world model and test whether it extrapolates the way that world model would. A model trained on orbital trajectories predicted them well, but symbolic regression on the adapted model recovered force laws that make no physical sense. This is our diagnostic D3. WM-Concord applies it to the same zoo as the other diagnostics, so its verdict can be compared model by model.

### 3.4. Probing confounds

Hewitt and Liang [7] pointed out that a high probe accuracy does not show that the model encodes a property, because an expressive probe can learn the task by itself. They proposed control tasks: the same inputs with labels assigned at random to each input type, so the labels carry no structure from the model. Selectivity, the probe's accuracy on the real task minus its accuracy on the control task, measures how much of the accuracy comes from the model rather than from the probe. They found that larger probes, such as multilayer perceptrons, often reach high accuracy with low selectivity. WM-Concord reports selectivity, not raw accuracy, for every D4 probe, using their control task construction, and adds a layer 0 baseline because the token embedding alone already reads part of the Othello board.

Belinkov [8] reviewed the probing classifier method and its limits. A probe shows that information can be decoded from a representation, not that the model uses it; results depend on the choice of probe and on the baseline it is compared with; and the correlation between probe accuracy and model behaviour is not a causal link. The review points to controls, simpler probes, and interventions on the representation as ways to close these gaps. WM-Concord follows this advice for D4 with a control task, a linear probe as the default, and a causal check, and records models whose state is decodable but not used as a separate outcome.

### 3.5. Measurement validity in ML evaluation

Jacobs and Wallach [9] brought measurement modelling from the social sciences to computational systems. An unobservable theoretical construct, such as fairness, is only ever measured through an operationalization, and every operationalization makes assumptions that must be checked. They describe how to do so through reliability and several kinds of validity, including convergent validity (do different measures of the same construct agree?) and predictive validity (does the measure predict outcomes it should predict?). WM-Concord treats "world model" as such a construct and the four diagnostics as its competing operationalizations: H1 and H2 test their convergent validity, and H3 tests their predictive validity against the fragility suite.

Raji et al. [10] argued that popular machine learning benchmarks are often presented as measures of general ability while they test narrow, specific tasks, so a high score is read as evidence for a broad capability that the benchmark cannot establish. They call for benchmarks to state clearly what they measure and to be validated against that claim. World model diagnostics face the same risk: each paper takes a pass on its own diagnostic as evidence that the model has a world model. WM-Concord asks whether these diagnostics measure the same thing at all, and which of them is backed by an external outcome.

### 3.6. Inference when runs are not independent

Cameron, Gelbach and Miller [11] studied regression inference when observations come in groups that share unobserved influences, such as workers in the same state. Standard errors that treat every observation as independent are then too small, and the problem is worst when there are few groups. They show that bootstrap methods which resample whole groups give much more honest tests, and that some refinements of them stay reliable even when there are few groups. Bouthillier et al. [12] measured how much the results of machine learning benchmarks move when only the random seed, the data order or the initialisation changes, and found that this variation is often as large as the differences papers report between methods; they recommend running several seeds and building that variation into every comparison. WM-Concord trains three seeds of every configuration, so its 81 Othello models form 27 groups of near copies. Following both works, every interval in this report resamples whole configurations, and H2 judges disagreement against the spread between seeds of the same configuration.

### 3.7. How WM-Concord differs

These papers reach opposite conclusions about the same domain, but they use different instruments on different models, so their disagreement cannot be interpreted. WM-Concord runs all four instruments on one shared zoo of models with a known ground truth, which turns the disagreement into something that can be measured (H1, H2) and judged against an external criterion (H3).

## 4. Data and Domains

All domains are defined by a deterministic finite automaton (DFA), so the true state is known at every step. Every domain emits the same triple: token sequence, true state, and the set of legal next tokens.

| Domain | State | Training distributions |
|---|---|---|
| Othello (done) | board position and player to move | synthetic, championship, mixed |
| Navigation (next) | current intersection on a street graph | shortest paths, noisy shortest paths, random walks |
| Interpreter (held out) | registers and stack of a small machine | random, short structured, long structured |

Table 1: The three domains. Othello is complete; navigation and the interpreter follow the same pipeline.

### 4.1. Othello

**The game.** Othello is a two player board game on an $8 \times 8$ grid with discs that are black on one side and white on the other. It starts with four discs in the centre, two of each colour on a diagonal, and black moves first. A move places one disc of the mover's colour on an empty square so that, along at least one of the eight directions, a straight unbroken line of the opponent's discs is enclosed between the new disc and another disc of the mover's colour. Every enclosed disc is flipped to the mover's colour. A move that flips nothing is illegal. A player with no legal move must pass, and the game ends when neither player can move, usually when the board is full after 60 moves; the player with more discs wins.

**Why Othello is the test bed.** Three properties make Othello the standard domain for world model studies [1, 5]. First, the state is exact and small: the board and the player to move are known after every move, so every diagnostic can be scored against the truth rather than against a human judgement. Second, the board is hard to read from the moves: the move list says only where discs were placed, never which discs were flipped, so knowing which discs are *mine* and which are *theirs* requires replaying the rules, which is exactly the internal state a world model is supposed to track. Third, the legal moves depend on the whole board, so a model that only learns which squares tend to follow which cannot stay legal for a whole game. The published Othello GPTs also give us two reference models with known results to check our code against.

**The state and the tokens.** The Othello state consists of the $8 \times 8$ board together with the player to move. Following the vocabulary used by Othello GPT [1], tokens 1 to 60 represent the 60 board squares other than the four fixed starting squares, while token 0 is used for padding. Our own rules engine replays every game and provides the board and the legal moves after every move; it is the only copy of the rules, shared by training, every diagnostic and the tests, and it found no illegal game in 101,000 replayed synthetic games.

We use three training distributions. The synthetic distribution is the `othello_world` corpus from Li et al. [1], about 19 million games generated from random legal moves. The championship distribution is drawn from the WTHOR tournament archive and contains 137,548 human games from 1977 through 2025. After removing 41 duplicate games, no illegal games remain, and the data are split into 110,005 training games, 13,751 validation games, and 13,751 test games. The mixed distribution pools both.

Every model draws 1,280,000 training games (5,000 steps of 256 games), uniformly with replacement from its pool. The synthetic pool holds 1,280,000 games, so each is drawn about once. The championship pool is all 110,005 WTHOR training games, so each is drawn about 11.6 times. The mixed pool holds 640,000 synthetic games and the 110,005 championship games, so mixed training is about 85% synthetic and 15% championship by games, while its validation and test sets are 50% of each. The preregistration names this distribution "mixed (50/50)"; the models were trained this way before any diagnostic score existed, and we report the exact mix here.

The three splits have fixed roles throughout the project. The training split is used only for model training, the validation split is used for checks during training and for fitting diagnostics, and the test split is reserved exclusively for final diagnostic scores.

### 4.2. Navigation

Navigation follows Vafa et al. [3]: the state is the current intersection on a street graph of Manhattan, a token is a turn, and the three training distributions are shortest paths, noisy shortest paths and random walks. It is the next domain to be built; it reuses the zoo, the diagnostics and the analysis code of Othello unchanged, and its first step is a reproduction of the published state equivalence numbers on the released navigation checkpoints.

### 4.3. Interpreter

The interpreter is a small deterministic stack or register machine whose full state (registers and stack) is known after every instruction, so the size of its automaton can be tuned. It is the held out domain for H4: its models are trained and scored only after H1 to H3 are fixed on Othello and navigation.

### 4.4. Reproduction artifacts

Two published Othello GPT models [1], one trained on synthetic games and one on championship games, serve as anchors: every diagnostic scores them to check the code against published findings, but they never enter H1 to H4, because they were trained with a different budget and code. For navigation, the released checkpoints of Vafa et al. [3] play the same role.

## 5. Methodology

### 5.1. Model zoo

The zoo varies four axes on purpose, so that models differ in quality and the diagnostics have something to disagree about: 3 architectures $\times$ 3 scales $\times$ 3 training distributions $\times$ 3 seeds, which gives 81 models per domain and 243 over the three domains. The architectures are a GPT style transformer [13], a stacked LSTM [14] and Mamba [15], run with the plain PyTorch `mambapy` build because the official CUDA kernels did not compile on our instances. They remember the past in different ways (attention over the whole history, a fixed size recurrent state, a selective state space recurrence), which is exactly the axis along which H2 asks whether the diagnostics disagree. The scales are small (4 layers, width 128), medium (6 layers, width 256) and large (8 layers, width 512), from 0.48 to 25.3 million parameters.

Every model gets the same budget: 5,000 steps of 256 games, AdamW with learning rate $3 \times 10^{-4}$, 200 warmup steps and weight decay 0.01. **Admission rule:** a model enters H1 to H3 only if its legal move rate on 1,000 validation games is at least 0.50 (an untrained model scores about 0.15). This rate is also the model's *competence* in the analysis. The rule was fixed before any diagnostic score existed; models that fail are still scored and reported.

### 5.2. Training and testing procedure

**How a model predicts the next move.** A game is a list of moves, and each move is one token: the square that was played. A model reads the moves made so far and outputs one score for each of the 60 squares; its prediction for the next move is the square with the highest score. There is no sliding window: a game has at most 60 moves and every model reads up to 59 tokens, so the whole game always fits, the history only grows and nothing is ever dropped. The transformer sees all earlier moves at once through attention, with a causal mask so that position $t$ cannot look at later moves; the LSTM and Mamba read the moves one at a time and carry a memory forward. Passes are not tokens, as in Othello GPT: after a pass the same player simply moves again, and the model has to infer this from the board it has tracked.

**Training objective.** For a game with moves $m_1, \ldots, m_T$, the model receives $m_1, \ldots, m_{T-1}$ and is trained to predict $m_{t+1}$ at every position $t$ at once, with the cross entropy loss

$$\mathcal{L} = -\frac{1}{T-1} \sum_{t=1}^{T-1} \log p_\theta(m_{t+1} \mid m_1, \ldots, m_t) \,.$$

This is *teacher forcing*: at every position the model reads the real history, never its own earlier guesses. All 81 models share the budget given above and one script, `src/zoo/train.py`, and the 81 runs were split over the four instances by `src/shellscripts/train_slice.sh`. Each run writes the checkpoint and a training record with the final loss, the validation legal move rate and the SHA 256 hash of the checkpoint, which every later step checks. Figure 1 shows the whole pipeline.

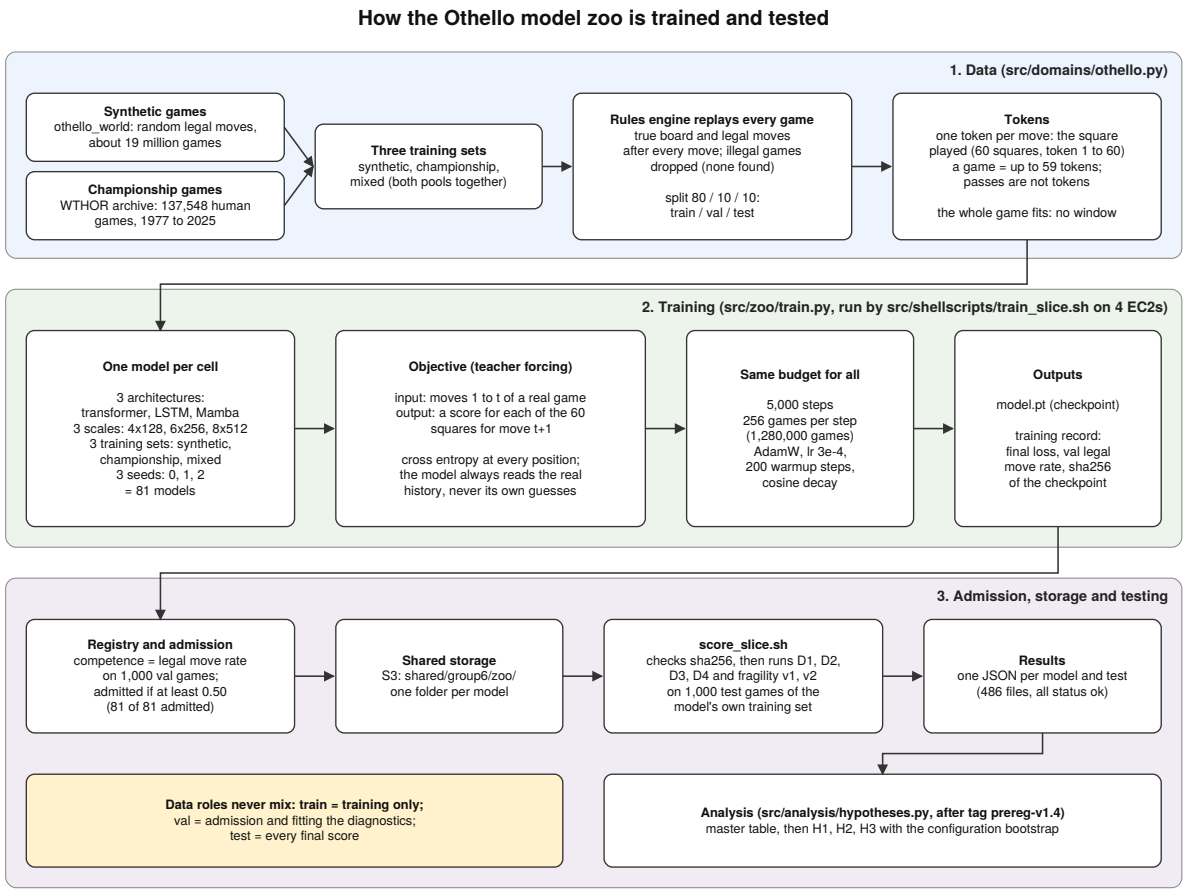



Figure 1: How the Othello zoo is trained and tested, from the game data to the analysis. Source: `reports/diagrams/zoo_training_pipeline.drawio`.

**How each split is used.** The split decides what a number may be used for, and no step reads another step's split (Table 2).

| Split | Used for |
|---|---|
| train | model training only: 5,000 steps of 256 games, 1,280,000 games drawn in all |
| val | admission (legal move rate on 1,000 games); fitting the diagnostics: D3 adapts on 64 positions, D4 fits its probes on 10,000 games and chooses the layer on 1,000 more |
| test | every final score: D1 to D4 and both fragility suites on 1,000 test games of the model's own training distribution, data seed 0 |

Table 2: The 80/10/10 split and what each part is used for. Synthetic test games are the last 10 of the 48 files of the `othello_world` validation folder; the championship split is a fixed random 80/10/10 split of the deduplicated WTHOR games (seed 0); mixed takes half of each.

**Training results.** Table 3 gives the outcome of training for every configuration. Competence rises with scale for every architecture and training set. The LSTMs are well behind the other two families at every scale. Every model has the same 5,000 step budget, so part of this gap may come from the budget rather than the architecture; we have not trained the LSTMs for longer, so this is untested (Section 8.1). The championship training loss is much lower than the

others, partly because its pool is small and each game is drawn about 11.6 times, so losses are comparable within a column but not across training sets.

| | | | Final training loss | | | Competence (val legal rate) | | |
|---|---|---|---|---|---|---|---|---|
| Architecture | Scale | Parameters | synthetic | champ. | mixed | synthetic | champ. | mixed |
| Transformer | small | 0.82M | 2.634 | 1.833 | 2.559 | 0.842 | 0.841 | 0.839 |
| Transformer | medium | 4.79M | 2.468 | 1.550 | 2.387 | 0.943 | 0.887 | 0.916 |
| Transformer | large | 25.31M | 2.358 | 1.094 | 2.258 | 0.969 | 0.899 | 0.928 |
| LSTM | small | 0.54M | 3.472 | 2.722 | 3.386 | 0.566 | 0.601 | 0.581 |
| LSTM | medium | 3.19M | 3.296 | 2.377 | 3.235 | 0.677 | 0.722 | 0.664 |
| LSTM | large | 16.87M | 3.181 | 2.181 | 2.886 | 0.754 | 0.772 | 0.809 |
| Mamba | small | 0.48M | 2.720 | 1.923 | 2.649 | 0.843 | 0.827 | 0.833 |
| Mamba | medium | 2.66M | 2.525 | 1.631 | 2.437 | 0.925 | 0.871 | 0.894 |
| Mamba | large | 13.63M | 2.384 | 1.131 | 2.281 | 0.963 | 0.887 | 0.924 |

Table 3: Training results, mean over 3 seeds per cell. Every model ran exactly 5,000 steps. Final loss is the mean training loss of the last 20 steps; competence is the legal move rate on 1,000 validation games of the model's own training distribution. Source: `results/zoo_training/`.

### 5.3. The four diagnostics

Every diagnostic implements one interface, `score(model, domain)`, and writes one JSON result per model with a single `score` for the master table, where higher always means more world model like, plus all details and the commit, GPU and library versions that produced it. Final scores use 1,000 test games of the model's own training distribution. Every protocol choice below was fixed in the preregistration and tuned, if at all, only on the two reference Othello GPTs, never on a zoo model.

**D1, next token validity.** The share of positions at which the model's top predicted move is legal. Nothing is fitted. A sampled rate (10 samples per position at temperature 1) is stored as a secondary value.

**D2, state equivalence.** Following the Myhill Nerode theorem [16] and Vafa et al. [3], two histories are the same state exactly when every continuation is equally allowed after both; we use the next move as the continuation. For a history $h$ the model's accepted set $A(h)$ is its $k$ best squares, where $k$ is the number of truly legal moves, so no threshold is tuned. The *compression error* of two histories that reach the same board is $|A(h) \triangle A(g)|/|A(h) \cup A(g)|$. The *distinction error* of two histories of the same length with different legal move sets is the share of boundary moves (legal after exactly one of them) that the model does not separate correctly. Up to 2,000 pairs of each kind are sampled, so each error has a standard error of at most 0.011. The score is one minus the mean of the two errors.

**D3, inductive bias probe.** Following Vafa et al. [4], a copy of the model gets a new linear head and is adapted on only 64 labelled validation positions to four yes or no questions of the form "is square $X$ mine?" ($X$ = C4, F5, C3, F6; mine means the disc belongs to the player who just moved). We ask "mine" and not "occupied" because occupancy can be read straight off the move list. The head learns at $10^{-3}$ and the body at $10^{-4}$ for 300 steps. A model of the same architecture and size with random weights is adapted in exactly the same way. The score is the mean gap in balanced accuracy between the two on 1,000 test positions, over 4 tasks and 3 seeds. A symbolic regression arm fits short formulas with PySR [17] from the true board to each adapted model's answer and records whether the formula recovers the "is $X$ mine" rule or

only an *occupancy shortcut* (a formula that reacts to $X$ being occupied but cannot tell mine from theirs).

**D4, representation probe.** After every layer, a linear probe reads each of the 64 squares as mine, theirs or empty [5]. Following Hewitt and Liang [7], an identical control probe learns random labels that depend only on the square just played and the position in the game. *Selectivity* is probe accuracy minus control accuracy. Probes are fitted on 10,000 validation games, the layer with the highest selectivity is chosen on 1,000 further validation games, and the score is that layer's selectivity on the test games. An MLP probe (one hidden layer of 256 units) is reported as a secondary result, together with an occupancy only baseline. A *causal check* after Li et al. [1] then edits the activations from the middle layer to the last so that the probes read one disc flipped, and compares the model's predictions on the moves whose legality changes with those after a random edit of the same size. A model with selectivity of at least 0.10 is *decodable*; it is *decodable and causal* if its causal score is at least 0.20 with an interval above 0, and *decodable but not causal* otherwise.

### 5.4. The four diagnostics in one box each

Each box gives what the test reads, what it computes and the range of its score. Notation: a game is a move list $m_1, \ldots, m_T$; $x_{\le t} = (m_1, \ldots, m_t)$ is the history after $t$ moves; $L(x_{\le t})$ is the set of truly legal next moves, given by our rules engine; $f$ is the model and $f(x_{\le t}) \in \mathbb{R}^{60}$ its scores for the next move. Every final score uses the 1,000 test games of the model's own training distribution.

**D1, next token validity: are the model's moves allowed?**

*Input:* the model $f$; every position $(g, t)$ of the test games that has a recorded next move, $N$ in all.
*Computation:* the top move $\hat{m}_{g,t} = \arg\max_v f(x_{g,\le t})_v$ is checked against the rules:

$$\mathrm{D1} = \frac{1}{N} \sum_{g,t} \mathbf{1}[\hat{m}_{g,t} \in L(x_{g,\le t})] \,.$$

*Output:* a rate in $[0, 1]$; 1 means every top move is legal. An untrained model scores about 0.15. Nothing is fitted.

**D2, state equivalence: two roads to the same board**

*Input:* histories $h$ (prefixes of test games); the accepted set $A(h)$, the model's $k$ highest scoring squares with $k = |L(h)|$; up to 2,000 *same state* pairs $S$ (different move lists, same board and player to move) and up to 2,000 *different state* pairs $D$ (same length, different legal sets).
*Computation:*

$$\mathrm{CE} = \operatorname*{mean}_{(h,g)\in S} \frac{|A(h) \triangle A(g)|}{|A(h) \cup A(g)|}, \qquad \mathrm{DE} = \operatorname*{mean}_{(h,g)\in D} \left(1 - \frac{|B_h \cap A(h) \setminus A(g)| + |B_g \cap A(g) \setminus A(h)|}{|B_h| + |B_g|}\right),$$

where $B_h = L(h) \setminus L(g)$ and $B_g = L(g) \setminus L(h)$ are the moves legal after only one of the two histories (the boundary). $\mathrm{D2} = 1 - \frac{1}{2}(\mathrm{CE} + \mathrm{DE})$.
*Output:* $[0, 1]$; 1 means the model treats equal boards alike and separates different ones exactly on their boundary. CE is the compression error, DE the distinction error.

**D3, inductive bias probe: does the model learn a new board question fast?**

*Input:* the model $f$; a copy $f^0$ of the same architecture and size with random weights; four tasks $j$ "is square $X$ mine?", $X \in \{\text{C4}, \text{F5}, \text{C3}, \text{F6}\}$; 64 labelled validation positions; seeds $s \in \{0, 1, 2\}$.

*Computation:* a copy of $f$ with a new linear head is adapted on the 64 positions (300 AdamW steps of 16, head learning rate $10^{-3}$, body $10^{-4}$), giving $f_{j,s}$; $f^0$ is adapted the same way, giving $f^0_{j,s}$. With balanced accuracy BA $= \frac{1}{2}$(hit rate on mine + hit rate on not mine) on 1,000 test positions,

$$\text{D3} = \frac{1}{12} \sum_{j=1}^{4} \sum_{s=0}^{2} \left[ \text{BA}(f_{j,s}) - \text{BA}(f^0_{j,s}) \right].$$

*Output:* a gap in $[-1, 1]$; 0 means pretraining gives no help on board questions. Zoo range $-0.01$ to 0.10, Othello GPT 0.14. A symbolic regression arm (PySR) records whether the adapted model follows the true rule or the occupancy shortcut; it does not change the score.

**D4, representation probe: can the board be read from inside the model, and does the model use it?**

*Input:* the vector $a_\ell(x_{\le t}) \in \mathbb{R}^d$ after each layer $\ell$, standardized per feature; the true label $y_{t,s} \in \{\text{empty}, \text{mine}, \text{theirs}\}$ of each of the 64 squares $s$; a control label $c_{t,s}$, one fixed random label per (type of position, square), where the type is the square just played and the move number.

*Computation:* a linear probe $P_\ell : \mathbb{R}^d \to \mathbb{R}^{64 \times 3}$ is trained with cross entropy on 10,000 validation games, and a control probe $Q_\ell$ the same way on $c$. With Acc the share of (position, square) labels predicted correctly,

$$\text{Sel}_\ell = \text{Acc}(P_\ell) - \text{Acc}(Q_\ell), \qquad \ell^* = \arg\max_\ell \text{Sel}_\ell, \qquad \text{D4} = \text{Sel}_{\ell^*} \text{ on the test games},$$

where $\ell^*$ is chosen on 1,000 further validation games. *Causal check:* the activations from the middle layer to the last are edited until the probes read one flipped disc with probability 0.99; causal score = share of boundary moves the model then gets right minus the same share after a random edit of equal size.

*Output:* selectivity in $[-1, 1]$, well below 1 in practice because the control probe also scores well (zoo 0.11 to 0.31, Othello GPT 0.44); decodable if Sel $\ge 0.10$; decodable and causal if the causal score is at least 0.20 with its interval above 0.

### 5.5. Fragility suite

H3 needs an outcome that no diagnostic could have been tuned toward. The fragility suite keeps the rules of Othello fixed and shifts the situations, so the right answer always comes from the true rules (changing the rules would require adaptation, which is what D3 measures). It has five kinds of variant: the model's own test games (the reference rate); games from the other training distributions; positions right after a pass, when the same player moves twice; the same games rotated by 180 degrees or reflected along either diagonal (the only symmetries that keep the start position); and self play with forced detours, where with probability $p \in \{0.1, 0.5\}$ the model's move is replaced by a random legal move or by the one it rates least likely. Fragility is the mean drop in legal move rate over the variants, and *relative fragility* divides it by the model's own in distribution rate, so models of different competence are comparable.

**Fragility v2: how much worse does the model get when the situation shifts?**

*Input:* the model's legal top move rate $r_0$ on its own test games, and the same rate $r_v$ under each variant $v$: the other training distribution(s) (one for synthetic and championship models, two for mixed), three board symmetries (rotation by 180 degrees and the two diagonal reflections), positions right after a pass, and four detour settings (random or adversarial replacement with $p \in \{0.1, 0.5\}$).

*Computation:* the drop of each variant is $\delta_v = r_0 - r_v$ (after a pass: the matched baseline minus the rate; detours: the self play rate at $p = 0$ minus the rate at $p$), and

$$F = \frac{1}{|V|} \sum_{v \in V} \delta_v, \qquad \text{relative fragility} = \frac{F}{r_0}.$$

*Output:* higher means more fragile; a negative value means the model does slightly better under the shifts than on its own games. Zoo range of relative fragility v2: $-0.124$ to $0.215$.

Version 1 was frozen with the git tag `fragility-v1` on October 6. Before the zoo was scored we found that two of its parts could not measure fragility on this zoo: its detours stopped at the first illegal proposal, so almost every model completed none of them, and it compared positions after a pass with all positions, while pass positions have many more legal squares. Version 2 (tag `fragility-v2`, October 18) plays every detour game to the end and scores the legal rate over all predictions, and compares each pass position with positions at the same move number and with the same number of legal moves. On the test model this matching moved the after pass drop from $-0.130$ to $-0.009$. H3 uses relative fragility v2; version 1 is reported as the preregistered secondary outcome.

### 5.6. Next move or whole game?

D1, D2, the probes of D4 and most of the fragility suite are computed with teacher forcing: at every position the model reads the real history of a test game and is judged on its next move only. This is the unit the preregistration fixes, for two reasons. It is the quantity every diagnostic in the literature is defined on, so our scores stay comparable with the published ones, and it separates the diagnostics from one another: a model that makes one early mistake in a game it plays itself reaches boards no test game contains, and every later score would then measure the mistake rather than the model. What it misses is that a model can be legal at almost every single step and still lose track of the board over a long game, because under teacher forcing an error never enters the history the model reads next.

To answer this directly we add two whole game measures, decided after the Othello results were read and therefore reported as exploratory (`src/analysis/sequence_othello.py`). *Legality by move number* splits the D1 rate by the move being predicted, on the same test games and positions, so its overall value equals D1 exactly. *Free play* lets the model play 1,000 whole games against itself from its own moves (the first move is random): its top move is checked against the rules at every step, and the move actually played is drawn from the model's own probabilities over the legal moves, so every game continues to its natural end and the games differ from one another. (A first version played the top legal move instead; because that choice is deterministic, its 1,000 games collapsed into at most four distinct games, one per opening move, and it was replaced before any result was used.) We report the legal rate per move number, the share of games with no illegal top move at all (the full game rate), and the move at which the first error happens. Both measures will be preregistered for navigation and the interpreter before any of their models is scored.

### 5.7. Analysis plan and preregistration

Every choice that could be tuned after seeing results was written down and frozen with a git tag before the results it affects existed (Table 4), and the diagnostic code refuses to score a zoo model until the required tag exists.

| Date | Tag | Content |
|---|---|---|
| Oct 6 | `fragility-v1` | fragility suite frozen before any score |
| Oct 6 | `prereg-v1` | admission rule, one score per diagnostic, H1 to H4 decision rules |
| Oct 13 | `prereg-v1.1` | Amendment 1: competence control, relative fragility, D2 to D4 protocols |
| Oct 16 | `prereg-v1.2` | Amendment 2: D4 MLP probe and causal check, D3 symbolic regression, D2 sample size |
| Oct 18 | `prereg-v1.3`, `fragility-v2` | Amendment 3: fragility v2 and the exact analysis details |
| Oct 18 to 19 | (scoring) | all 81 models scored by D1 to D4 and both fragility suites |
| Oct 19 | `prereg-v1.4` | Amendment 4: configuration bootstrap, H3 without the test model, fragility distribution first |

Table 4: Preregistration timeline. Amendment 4 followed Instructor Review 5; it was written after the zoo was scored but before any H1 to H3 statistic was computed, and it changes no score.

**Intervals.** The 81 models form 27 configurations of three seeds each, and seeds of one configuration are near copies. Every interval is therefore a 95% percentile cluster bootstrap [11]: each of 2,000 resamples (seed 0) draws 27 configurations with replacement and keeps all seeds of each.

**H1.** For each of the six pairs of diagnostics we compute Kendall's tau [18] (tau b, which handles ties) and Spearman's rho, and Kendall's W [19] across all four. A pair *agrees* if tau is at least 0.50 and its interval excludes 0, *disagrees* if its interval includes 0 or tau is below 0.30, and is *partial* otherwise. Because every diagnostic may simply follow competence, each pair is also tested with Kendall's partial tau [20] given competence,

$$\tau_{xy\cdot c} = \frac{\tau_{xy} - \tau_{xc}\,\tau_{yc}}{\sqrt{(1-\tau_{xc}^2)(1-\tau_{yc}^2)}},$$

and W is recomputed on ranks with competence regressed out. A pair that agrees raw but disagrees after the control is reported as *agreement through competence*, and the controlled result is the one used to say whether the diagnostics measure the same thing. If tau(D1, competence) is at least 0.90, the controlled D1 pairs are marked not interpretable (rule A5). As a sensitivity check (A3), H1 is repeated on the models with competence of at least 0.90.

**H2.** Each model's disagreement is the standard deviation of its four rank percentiles. It is regressed on architecture, scale and training distribution; disagreement is *systematic* if a factor is significant at 0.05 after Holm correction and its effect exceeds the seed noise floor (the spread of disagreement across seeds of one configuration).

**H3.** The outcome is relative fragility v2, whose distribution is reported first. D1 is the reference: we report its Spearman correlation with the outcome. For D2, D3 and D4 we compute the partial Spearman correlation with the outcome given D1, and a least squares regression of the outcome on D1, the diagnostic, architecture and scale. A diagnostic *adds to D1* if its partial correlation interval excludes 0, and is the *best predictor* if its partial correlation is the strongest and the interval of each difference to the other two excludes 0. The analysis is also reported without the model whose fragility v1 output motivated Amendment 3, and on fragility v1. The D4 causal label is tested for predictive value beyond D4 selectivity.

**H4.** H1 to H3 are fixed on Othello and navigation before any interpreter model is scored; the findings generalize if every pair keeps its H1 category and the H3 verdict is the same on the interpreter.

**Exploratory analyses.** Analyses decided after the preregistered results were read are labelled exploratory, use the same bootstrap, and never replace a preregistered result.

## 6. Experimental Setup

Training and scoring ran on four AWS g5.2xlarge instances (one NVIDIA A10G with 24 GB, 8 vCPUs, 32 GB RAM each), one per team member, each with one slice of about 20 models. Library versions are pinned in `requirements.txt`. Every training record stores the checkpoint's SHA 256 hash, and the scoring launcher (`src/shellscripts/score_slice.sh`) checks it before scoring, skips results that already exist and resumes after an interrupted instance. Every result records the commit that produced it; 137 test functions run in GitHub Actions on every pull request. Measured training time per model at the full budget (Table 5) puts one domain at about 37 GPU hours, 88% of it Mamba in plain PyTorch. Scoring one model with all four diagnostics and both fragility suites takes minutes, dominated by D3, which adapts 24 copies of the model.

| Architecture | small ($4 \times 128$) | medium ($6 \times 256$) | large ($8 \times 512$) |
|---|---|---|---|
| Transformer | 1.5 min | 4.7 min | 16.1 min |
| LSTM | 0.6 min | 1.6 min | 5.9 min |
| Mamba (`mambapy`) | 17.7 min | 53.0 min | 145.7 min |

Table 5: Measured training time per model on one A10G at 5,000 steps of 256 games.

## 7. Results

All results in this section are for Othello. Numbers in brackets are 95% configuration bootstrap intervals.

### 7.1. Reproduction of published results

Before building on our code we reproduced a published number. Linear probes on the published synthetic Othello GPT match Nanda et al. [5] within one percentage point, a tolerance fixed before comparing, on five of six values (Table 6). The key finding reproduces: the board is almost perfectly linear in mine and theirs labels, but only about 75% readable in black and white labels. The one miss, black and white at layer 0, most likely reflects a different definition of layer 0 and does not affect D4, which uses mine and theirs labels. The model's top move is legal 99.93% of the time on our test games.

| Labels | Layer | Ours (%) | Nanda et al. (%) | Difference |
|---|---|---|---|---|
| mine, theirs, empty | 0 | 90.5 | 90.9 | −0.4 |
| mine, theirs, empty | 4 | 98.7 | 99.0 | −0.3 |
| mine, theirs, empty | 7 | 98.9 | 99.5 | −0.6 |
| black, white, empty | 0 | 75.8 | 62.2 | +13.6 |
| black, white, empty | 4 | 75.9 | 75.0 | +0.9 |
| black, white, empty | 7 | 75.2 | 74.4 | +0.8 |

Table 6: Reproduction gate: linear probe accuracy on the synthetic Othello GPT.

All four diagnostics also order the two reference Othello GPTs as the literature does (Table 7). Our D2 gives the same order as Vafa et al. [3], and only the synthetic model passes the causal check, while the championship model's board is readable but barely used.

| Diagnostic | Othello GPT synthetic | Othello GPT championship |
|---|---|---|
| D1 next token validity | 0.9996 | 0.896 |
| D2 state equivalence | 0.999 | 0.766 |
| D3 inductive bias gap | 0.140 | 0.059 |
| D4 probe selectivity | 0.435, decodable and causal | 0.272, decodable but not causal |

Table 7: Anchor scores on the test split. The anchors check the code and never enter H1 to H4.

### 7.2. The Othello zoo

All 81 models were trained and all 81 passed the admission rule; competence ranges from 0.559 to 0.969 (mean 0.821). All 486 result files (81 models × four diagnostics and two fragility suites) have status ok, and none is missing. Table 8 gives the mean of every quantity per architecture and scale. Every diagnostic rises with scale in every family; the LSTMs are the weakest on all four, while transformers and Mamba are close. All ten models that pass the causal check are medium or large transformers or Mamba, and nine of the ten were trained on synthetic games. Twenty three models have competence of at least 0.90.

| Architecture | Scale | Competence | D1 | D2 | D3 | D4 | Rel. fragility v2 | Causal |
|---|---|---|---|---|---|---|---|---|
| Transformer | small | 0.841 | 0.843 | 0.751 | 0.014 | 0.194 | 0.022 | 0 of 9 |
| Transformer | medium | 0.915 | 0.917 | 0.817 | 0.038 | 0.247 | 0.021 | 3 of 9 |
| Transformer | large | 0.932 | 0.933 | 0.840 | 0.056 | 0.268 | 0.013 | 4 of 9 |
| LSTM | small | 0.583 | 0.584 | 0.607 | 0.013 | 0.120 | 0.008 | 0 of 9 |
| LSTM | medium | 0.688 | 0.690 | 0.632 | 0.014 | 0.155 | 0.020 | 0 of 9 |
| LSTM | large | 0.778 | 0.780 | 0.666 | 0.017 | 0.174 | 0.014 | 0 of 9 |
| Mamba | small | 0.835 | 0.837 | 0.722 | 0.025 | 0.203 | 0.019 | 0 of 9 |
| Mamba | medium | 0.897 | 0.899 | 0.789 | 0.044 | 0.240 | 0.007 | 1 of 9 |
| Mamba | large | 0.924 | 0.925 | 0.826 | 0.070 | 0.269 | 0.008 | 2 of 9 |

Table 8: Means over the 9 models (3 distributions × 3 seeds) of each architecture and scale. Causal: models labelled decodable and causal.

**Inside each diagnostic.** D1 is almost the same quantity as competence (tau 0.968). For D2, models treat histories that reach the same board alike fairly well (mean compression error 0.146, range 0.021 to 0.289) but often fail to separate boards whose legal moves differ (distinction error 0.376, range 0.160 to 0.648). For D3, the adapted pretrained models reach a balanced accuracy of 0.579 against 0.547 for random weight models, a mean gap of 0.032, well below the anchors. The symbolic regression recovers the "is $X$ mine" rule in only 3.8% of searches (2.7% for random weight models), while the occupancy shortcut appears in 22.7% (14.8%), and across models the D3 score rises with the shortcut rate (Spearman 0.357 [0.041, 0.605]): part of what D3 rewards is the shortcut. For D4, the mean linear selectivity is 0.208 (0.111 to 0.314) and the MLP selectivity is lower, 0.183, as Hewitt and Liang predict. The best layer sits at 83% of the depth on average. Every model is decodable, but only 10 of 81 pass the causal check (mean causal score 0.126).

## 7.3. Concordance (H1)

| Pair | Kendall tau | Raw | Partial tau given competence | Controlled |
|---|---|---|---|---|
| D1, D2 | 0.842 [0.711, 0.920] | agree | 0.303 [0.154, 0.539] | not interpretable (A5) |
| D1, D3 | 0.574 [0.407, 0.688] | agree | 0.076 [−0.003, 0.211] | not interpretable (A5) |
| D1, D4 | 0.801 [0.688, 0.867] | agree | 0.238 [0.106, 0.440] | not interpretable (A5) |
| D2, D3 | 0.480 [0.264, 0.631] | partial | 0.006 [−0.107, 0.171] | disagree |
| D2, D4 | 0.683 [0.504, 0.804] | agree | 0.090 [−0.029, 0.341] | through competence |
| D3, D4 | 0.553 [0.376, 0.680] | agree | 0.194 [0.003, 0.393] | through competence |
| Kendall W | 0.872 [0.764, 0.921] | | 0.267 [0.189, 0.427] | on residual ranks |

Table 9: H1 on 81 models. "Through competence": agreement through competence.

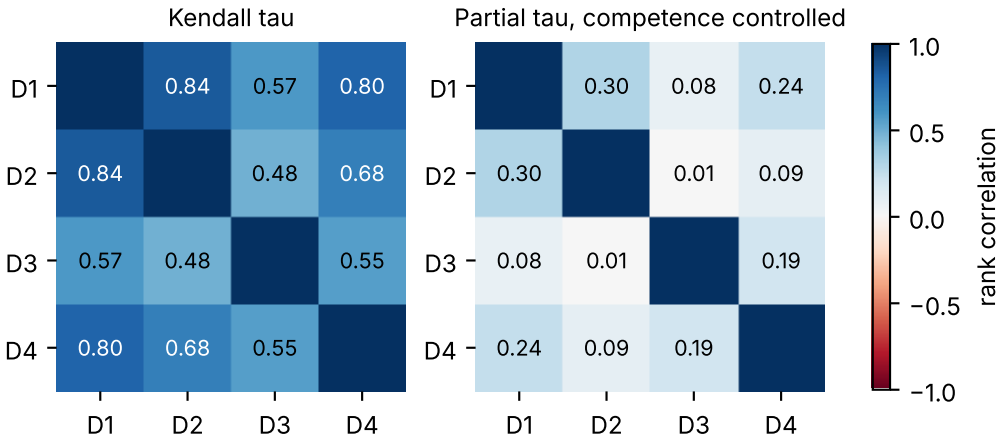



Figure 2: Rank agreement between the diagnostics, raw (left) and with competence controlled (right).

On raw scores the diagnostics agree: five of six pairs meet the agreement rule, D2 with D3 is partial, and Kendall's W is 0.872 (Table 9, Figure 2). But every diagnostic follows competence: its tau with competence is 0.968 for D1, 0.826 for D2, 0.790 for D4 and 0.577 for D3. D1 is above the 0.90 line of rule A5, so its controlled pairs are not interpretable, and the controlled conclusion rests on the three pairs among D2, D3 and D4. All three lose their agreement: D2 with D3 disagrees outright, and D2 with D4 and D3 with D4 are agreement through competence. On ranks with competence removed, W falls to 0.267.

**Sensitivity (A3).** On the 23 models with competence of at least 0.90, where tau(D1, competence) is 0.855 and A5 does not apply, three pairs still agree on raw scores (D1 with D2, D1 with D4, D2 with D4), and after the control only D1 with D2 agrees (partial tau 0.532 [0.088, 0.838]); W is 0.830 raw and 0.466 on residual ranks. The direction is the same as on the full zoo.

**Answer to H1.** The four diagnostics agree on which Othello models are better, but mostly because they all measure how competent the model is. With competence controlled, the three diagnostics that can be separated from competence do not measure the same thing.

## 7.4. Disagreement (H2)

The mean disagreement per model (the standard deviation of its four rank percentiles) is 0.089, against a seed noise floor of 0.028. Table 10 gives the mean disagreement per level of each factor

and the F test of each factor in one regression with all three. No factor is significant after the Holm correction, so under the preregistered rule **the disagreement is consistent with noise**. Scale comes closest: small models are ranked least consistently.

| Factor | Levels | Mean per level | Effect | $F$ | $p$ | Holm $p$ | Systematic |
|---|---|---|---|---|---|---|---|
| Architecture | transformer, LSTM, Mamba | 0.094, 0.087, 0.084 | 0.010 | 0.25 | 0.783 | 0.783 | no |
| Scale | small, medium, large | 0.110, 0.083, 0.073 | 0.038 | 3.79 | 0.027 | 0.081 | no |
| Training data | synthetic, champ., mixed | 0.075, 0.102, 0.089 | 0.027 | 1.81 | 0.171 | 0.341 | no |

Table 10: H2 on 81 models. Effect: largest minus smallest level mean. A factor is systematic only if it is significant at 0.05 after Holm and its effect exceeds the seed noise floor (0.028). $F$ tests have 2 and 74 degrees of freedom.

### 7.5. Predictive validity (H3)

**The outcome first.** Relative fragility v2 has mean 0.015, standard deviation 0.070 and range −0.124 to 0.215, and 52% of the models are below 0, that is, slightly better under the shifts than on their own test games. Most models are therefore not very fragile, and the spread H3 tries to predict is small.

| Diagnostic | Partial Spearman given D1 [95% CI] | Regression coefficient [95% CI] | Adds to D1 |
|---|---|---|---|
| D1 (reference) | Spearman 0.150 [−0.315, 0.588] | | |
| D2 | 0.031 [−0.297, 0.438] | −0.03 [−1.02, 0.95] | no |
| D3 | 0.026 [−0.373, 0.386] | −0.08 [−2.20, 2.24] | no |
| D4 | −0.554 [−0.734, −0.124] | −1.48 [−2.47, −0.60] | yes |

Table 11: H3 on 81 models, outcome relative fragility v2. The regression is least squares of the outcome on D1, the diagnostic, architecture and scale; the coefficient shown is the diagnostic's. Fragility is "higher breaks more", so a negative value means a higher score goes with less breaking. Best predictor rule: D4 is strongest, but its difference to D2 has the interval [−1.114, 0.098], so no single diagnostic wins.

D1 alone does not predict fragility. Among D2, D3 and D4, only D4 adds to D1 (Table 11); its regression coefficient with D1, architecture and scale is −1.48 [−2.47, −0.60]. D4 is the strongest, but its difference to D2 has the interval [−1.114, 0.098], which includes 0, so under the preregistered rule **no single diagnostic wins**. The result is the same without the test model (−0.561 [−0.738, −0.117]) and on relative fragility v1 (−0.540 [−0.693, −0.197]). The raw Spearman correlations of all four diagnostics with the outcome are near 0 (0.15, 0.15, 0.13 and −0.04), so the D4 signal appears only after D1 is controlled. The causal label adds nothing beyond D4 selectivity (partial −0.049 [−0.241, 0.122]).

### 7.6. Exploratory: where the fragility and the D4 signal come from

The analyses in this subsection were decided after the preregistered results were read and are reported as exploratory. The models trained on human championship games behave differently from the rest: their mean relative fragility is +0.090, against −0.024 for synthetic and −0.022 for mixed training, and their mean D4 is the lowest (0.175, against 0.236 for synthetic). Figure 3 shows why they are fragile: they break on random games (a mean drop of 0.190) and on rotated or reflected boards (0.14 to 0.18), while synthetic and mixed models barely change under any

shift. Human games cover a narrow part of the game tree and are recorded in one orientation. Under forced detours every group is slightly more often legal than on its own line, so detours do not make these models fragile.

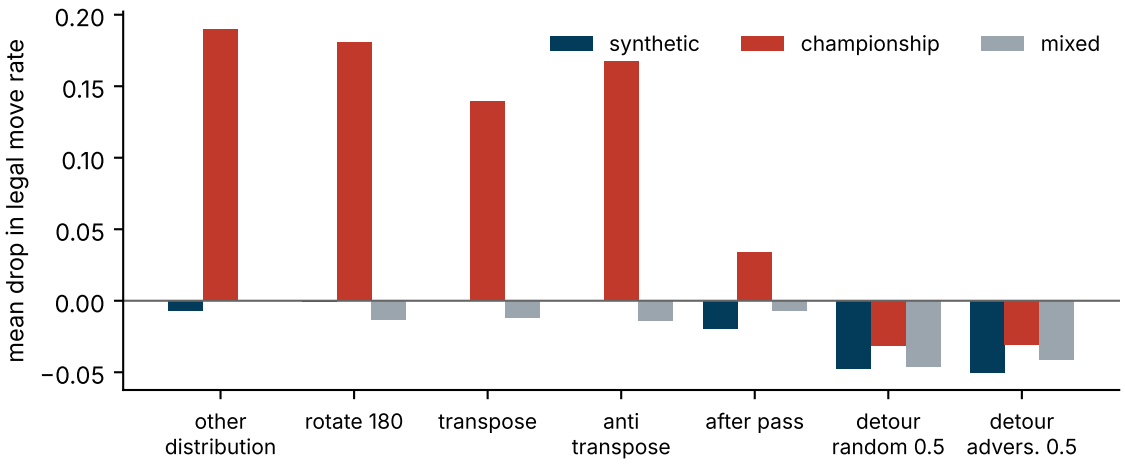



Figure 3: Mean drop in legal move rate per fragility v2 shift, by training distribution (exploratory). "Other distribution" averages both other distributions for mixed models.

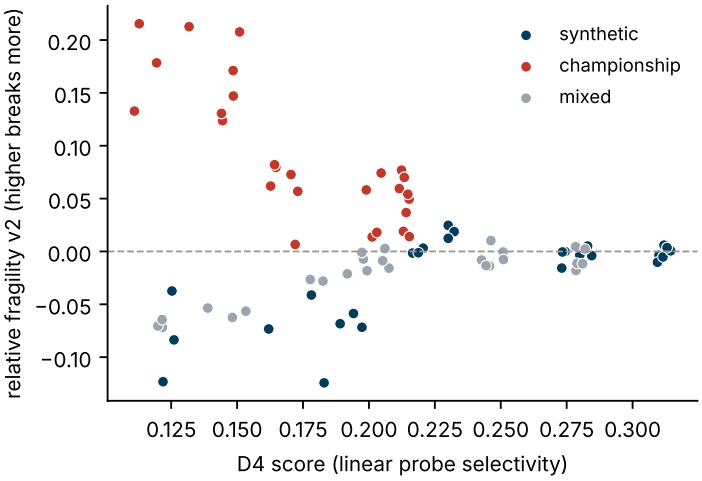



Figure 4: D4 against relative fragility v2, one point per model, by training distribution.

| D4 partial Spearman with relative fragility v2, given D1 | value [95% CI] |
|---|---|
| all 81 models (preregistered) | −0.554 [−0.734, −0.124] |
| within synthetic (27 models) | +0.449 [−0.226, 0.726] |
| within championship (27 models) | −0.530 [−0.761, 0.191] |
| within mixed (27 models) | +0.341 [−0.221, 0.666] |
| all 81, controlling for D1 and training distribution | +0.231 [−0.336, 0.589] |

Table 12: The D4 result within and across training distributions (exploratory).

Within each training distribution the D4 link is not there: its sign flips between distributions and every interval includes 0, and with training data added as a control it is +0.231 (Table 12, Figure 4). The same holds for D2 (+0.153 [−0.346, 0.555]) and D3 (−0.259 [−0.502, 0.078]). The

preregistered D4 result is computed exactly as planned, but it is a between distribution effect, a Simpson's paradox: it says that championship trained models are both less probe readable and more fragile, not that probing predicts fragility among comparable models. With nine configurations per distribution the within distribution estimates are imprecise, so the honest summary is that **no diagnostic shows a reliable link to fragility among comparable Othello models**.

**How much of H3 rests on the most fragile models?** The fragility that H3 predicts is concentrated in very few models: all 20 models with relative fragility above 0.05 were trained on championship games, and the nine most fragile configurations are exactly the nine championship configurations, led by the three LSTM ones (mean relative fragility 0.20, 0.18 and 0.13). Table 13 repeats the preregistered D4 statistic after leaving out the most fragile configurations. Leaving out any single configuration barely moves it (from −0.628 to −0.485), and leaving out the most fragile one keeps its interval below 0. Leaving out the three championship LSTM configurations already brings the interval across 0, and without all nine championship configurations the sign flips. The preregistered D4 result is therefore carried by the nine championship configurations, and mostly by the three championship LSTMs (exploratory).

| Models left out | Kept | D4 partial given D1 [95% CI] |
|---|---|---|
| none (preregistered) | 81 | −0.554 [−0.734, −0.124] |
| most fragile configuration (LSTM medium, championship) | 78 | −0.521 [−0.722, −0.074] |
| 3 most fragile (the three championship LSTMs) | 72 | −0.379 [−0.642, 0.075] |
| 5 most fragile | 66 | −0.243 [−0.560, 0.215] |
| 9 most fragile (every championship configuration) | 54 | +0.259 [−0.136, 0.505] |
| any one configuration, in turn | 78 | from −0.628 to −0.485 |

Table 13: Sensitivity of the H3 result to the most fragile configurations, ranked by mean relative fragility v2 over their three seeds (exploratory; configuration bootstrap as in Amendment 4).

### 7.7. Exploratory: next move versus whole game

All 81 zoo models and both anchors were measured as described in Section 5.6. The teacher forced rate reproduces each model's D1 score, except for one position out of about 58,000 in three models (LSTM large synthetic seed 1, Mamba large synthetic seed 1 and Mamba large mixed seed 1), a near tie in the top move that the GPU breaks differently at a different batch size. In free play, each model played at least 978 distinct games out of 1,000.

| | | Teacher forced | | Free play (own moves) | | | |
|---|---|---|---|---|---|---|---|
| Architecture | Scale | legal (D1) | full games | legal | full games | errors/game | 1st error |
| Transformer | small | 0.843 | 0.000 | 0.768 | 0.000 | 13.7 | 11.6 |
| Transformer | medium | 0.917 | 0.041 | 0.863 | 0.023 | 8.1 | 18.4 |
| Transformer | large | 0.933 | 0.116 | 0.888 | 0.111 | 6.6 | 22.9 |
| LSTM | small | 0.584 | 0.000 | 0.412 | 0.000 | 34.7 | 5.8 |
| LSTM | medium | 0.690 | 0.000 | 0.489 | 0.000 | 30.1 | 8.4 |
| LSTM | large | 0.780 | 0.000 | 0.639 | 0.000 | 21.3 | 11.0 |
| Mamba | small | 0.837 | 0.001 | 0.789 | 0.001 | 12.5 | 10.9 |
| Mamba | medium | 0.899 | 0.015 | 0.837 | 0.012 | 9.6 | 16.7 |
| Mamba | large | 0.925 | 0.079 | 0.864 | 0.075 | 8.0 | 22.4 |
| trained on synthetic | | 0.832 | 0.060 | 0.802 | 0.045 | 11.7 | 18.2 |
| trained on championship | | 0.814 | 0.003 | 0.572 | 0.000 | 25.2 | 7.2 |
| trained on mixed | | 0.823 | 0.021 | 0.808 | 0.029 | 11.3 | 17.3 |
| Othello GPT synthetic | | 0.9996 | 0.981 | 0.999 | 0.965 | 0.04 | 54.6 |
| Othello GPT championship | | 0.896 | 0.019 | 0.949 | 0.466 | 3.0 | 28.1 |

Table 14: Whole game measures (exploratory), means over 9 models per architecture and scale and over 27 per training distribution. Full games: share of games in which every top move is legal. First error: mean move number of the first illegal top move, over games that have one.

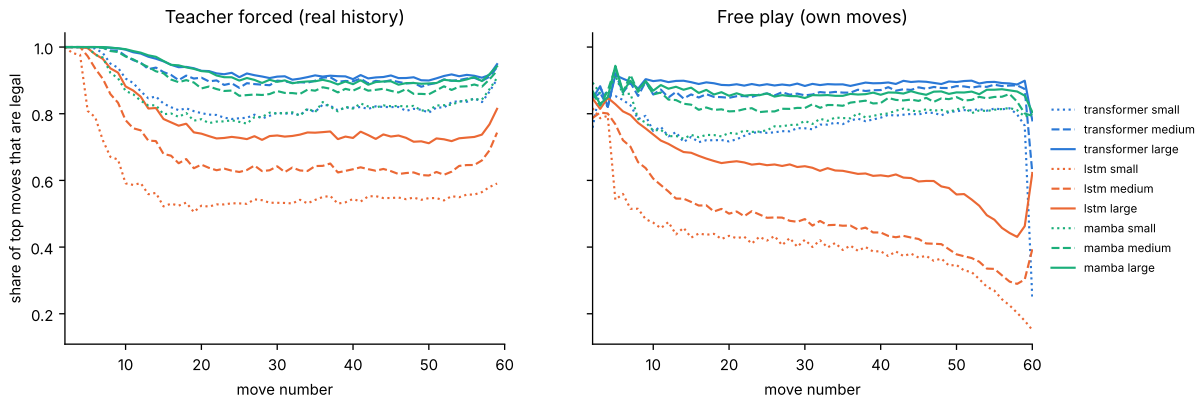



Figure 5: Share of top moves that are legal at each move number, with the real history (left) and in games the model plays itself (right); mean over the 9 models of each architecture and scale (exploratory).

Three things answer the question from the presentation (Table 14, Figure 5). First, **playing its own games is much harder than following real ones for the models trained on human games**: on average the free play rate is below the teacher forced rate by 0.03 for synthetic and 0.015 for mixed training (18 synthetic or mixed transformers and Mamba are even slightly higher in free play, by at most 0.024), but by 0.24 for championship training, and lower for all 27 championship models. Models trained on human games are legal 81% of the time on human test games but only 57% in their own games, and make their first illegal move at move 7 on average: as soon as the random opening takes them off the human lines, they fail, which is the same weakness the fragility suite found on random games and rotated boards. Second, **the LSTMs lose track of the board over a long game**: with the real history their legal rate is flat after the opening, but in their own games it falls steadily (for LSTM large, from 0.74 at move 10 to 0.64 at move 30 and 0.43 at move 58), while the transformers and Mamba stay flat, so the per move D1 score understates how much worse the LSTMs are at playing a whole game. Third, a

whole game without any illegal move is rare: besides the two anchors, only the large transformers trained on synthetic or mixed games and the large Mamba models trained on synthetic games reach it in more than 10% of games (at most 0.216 in the zoo, 0.965 for Othello GPT synthetic), and 41 of the 81 models never manage one, so this share sits at its floor for most of the zoo and the free play legal rate is the more informative number.

The whole game view ranks the models much as D1 does: the free play legal rate has Spearman 0.886 [0.721, 0.951] with D1 and 0.879 [0.710, 0.946] with competence, and neither whole game measure is related to relative fragility v2 (−0.179 [−0.599, 0.366] for the free play rate). With D1 controlled, D4 is the only diagnostic that relates to the full game rate (partial Spearman 0.354 [0.169, 0.526]; D2 0.022, D3 0.063, both intervals including 0), which fits the causal check: models whose board can be read and used play cleaner whole games. This is exploratory and is one of the measures we preregister for navigation. In short, whole game measures do not change the H1 to H3 answers, but they show two failures the next move unit misses: models trained on human games break once they leave human lines, and LSTMs drift over long games.

### 7.8. Held out domain (H4)

H4 is not yet tested. It requires H1 to H3 to be fixed on Othello and navigation first; the interpreter models are then trained and scored with the same code, and the Othello answers above are the first half of what they will be compared against.

## 8. Discussion

The Othello results answer the question that motivated the project, at least for one domain. Published claims for and against world models in Othello rely on different diagnostics, and our results suggest why they can disagree without either being wrong in its own terms: the diagnostics share a large common factor, competence, and beyond it they measure different things. A paper that reports one diagnostic on models of different quality will mostly be reporting how good the models are.

What each diagnostic licenses a researcher to claim follows from this. D1 measures competence and nothing more. D2 adds information about whether the model's predictions depend on the board rather than the move order, but its distinction error shows that even good models often fail to separate boards with different legal moves. D3, at the protocol we preregistered, gives small gaps on the zoo and partly rewards an occupancy shortcut that the move list gives away. D4 shows that the board is decodable in every model, while the causal check shows that only a few models, almost all trained on synthetic games, use it. "Decodable" is therefore not the same claim as "has a world model", which is what Li et al. [1] and Belinkov [8] warned about.

For predictive validity, the result is a caution rather than a ranking: on Othello the fragility suite mostly separates the championship trained models from the rest, so any diagnostic that also separates them looks predictive (Table 13). Navigation, where the three training distributions differ less sharply in coverage, is a better test of H3, and we plan to preregister a training distribution control for it before any navigation score exists.

### 8.1. Limitations

- **One domain so far.** All results are for Othello; navigation and the held out interpreter are still to come, and H4 is untested.
- **Few independent units.** The 81 models are 27 configurations; the configuration bootstrap makes the intervals honest but wide, and within distribution analyses rest on nine configurations each.

- **D2 uses next move continuations only**, which keeps it cheap but is a simplification of Vafa et al. [3], and its same board pairs come mostly from the opening.
- **D3 measures help from pretraining with 64 examples**, not closeness to a perfect answer, and part of it reflects an occupancy shortcut.
- **The detour reference of fragility v2.** The undisturbed self play run that each detour drop is measured against plays the model's best legal move after a random first move, so its games are deterministic and collapse into at most four distinct games; the detour drops therefore compare the detoured games with a very small reference. We found this after scoring, while building the free play measure. The preregistered suite is kept unchanged and reported as computed, and the navigation suite will draw its undisturbed games from the model's probabilities over the legal moves, as free play does.
- **Fragility is small on most models**, so H3 rests on the minority of fragile models; version 1 of the suite had two parts that could not measure fragility on this zoo, corrected in version 2 before scoring.
- **The mixed distribution** trains on about 85% synthetic games, not 50%, as described in Section 4.
- **Next move metrics.** D1 to D4 and most of the fragility suite judge the next move given the real history (teacher forcing), which is the preregistered unit; whole game measures are added for Othello as an exploratory check (Section 5.6) and preregistered for the next domains.
- **The LSTMs** are the weakest family at every scale under the shared budget of 5,000 steps; whether a longer budget or an LSTM with attention closes the gap is untested.
- **Compute.** Every model has the same modest budget on a single A10G; the best zoo models (competence up to 0.969) remain below the published Othello GPT, and Mamba runs without its optimized kernels.

## 9. Conclusion and Future Work

In one sentence: on Othello, the four world model diagnostics agree mainly because they all track competence, none of them reliably predicts when a model will break, and the one link to fragility comes from the training data. In detail, on 81 Othello models and under a preregistered analysis: **H1**, the four diagnostics agree on which models are better, but almost entirely through competence, and the diagnostics that can be separated from competence do not agree once it is controlled. **H2**, their disagreement is not systematic in architecture, scale or training data. **H3**, no single diagnostic wins; D4 adds to the legal move rate, but an exploratory check shows that this comes from differences between training distributions. **H4** is still to be tested on the held out interpreter.

Next, the same pipeline runs on navigation, starting with a reproduction of the state equivalence results of Vafa et al. [3], and then on the held out interpreter for H4, with the paper due on December 8. Beyond this project, the same comparison could be extended to domains with continuous or probabilistic state, and any new world model diagnostic can be added through the common interface and judged against these four on identical ground truth.

## References

[1] K. Li, A. K. Hopkins, D. Bau, F. Viégas, H. Pfister, and M. Wattenberg, "Emergent world representations: Exploring a sequence model trained on a synthetic task," in

*International Conference on Learning Representations (ICLR)*, 2023. [Online]. Available: https://openreview.net/forum?id=DeG07_TcZvT

[2] Y. Yuan and A. Søgaard, "Revisiting the othello world model hypothesis," in *ICLR 2025 Workshop on World Models*, 2025. [Online]. Available: https://arxiv.org/abs/2503.04421

[3] K. Vafa, J. Y. Chen, A. Rambachan, J. Kleinberg, and S. Mullainathan, "Evaluating the world model implicit in a generative model," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2024. [Online]. Available: https://arxiv.org/abs/2406.03689

[4] K. Vafa, P. G. Chang, A. Rambachan, and S. Mullainathan, "What has a foundation model found? using inductive bias to probe for world models," in *International Conference on Machine Learning (ICML)*, 2025. [Online]. Available: https://arxiv.org/abs/2507.06952

[5] N. Nanda, A. Lee, and M. Wattenberg, "Emergent linear representations in world models of self-supervised sequence models," in *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*, 2023. [Online]. Available: https://arxiv.org/abs/2309.00941

[6] S. Toshniwal, S. Wiseman, K. Livescu, and K. Gimpel, "Chess as a testbed for language model state tracking," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2022. [Online]. Available: https://arxiv.org/abs/2102.13249

[7] J. Hewitt and P. Liang, "Designing and interpreting probes with control tasks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing (EMNLP-IJCNLP)*, 2019. [Online]. Available: https://aclanthology.org/D19-1275

[8] Y. Belinkov, "Probing classifiers: Promises, shortcomings, and advances," *Computational Linguistics*, vol. 48, no. 1, pp. 207–219, 2022. [Online]. Available: https://aclanthology.org/2022.cl-1.7

[9] A. Z. Jacobs and H. Wallach, "Measurement and fairness," in *Proceedings of the 2021 ACM Conference on Fairness, Accountability, and Transparency (FAccT)*, 2021. [Online]. Available: https://arxiv.org/abs/1912.05511

[10] I. D. Raji, E. M. Bender, A. Paullada, E. Denton, and A. Hanna, "AI and the everything in the whole wide world benchmark," in *Advances in Neural Information Processing Systems, Datasets and Benchmarks Track*, 2021. [Online]. Available: https://arxiv.org/abs/2111.15366

[11] A. C. Cameron, J. B. Gelbach, and D. L. Miller, "Bootstrap-based improvements for inference with clustered errors," *The Review of Economics and Statistics*, vol. 90, no. 3, pp. 414–427, 2008.

[12] X. Bouthillier, P. Delaunay, M. Bronzi, A. Trofimov, B. Nichyporuk, J. Szeto, N. Sepah, E. Raff, K. Madan, V. Voleti, S. E. Kahou, V. Michalski, D. Serdyuk, T. Arbel, C. Pal, G. Varoquaux, and P. Vincent, "Accounting for variance in machine learning benchmarks," in *Proceedings of Machine Learning and Systems (MLSys)*, vol. 3, 2021. [Online]. Available: https://arxiv.org/abs/2103.03098

[13] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. [Online]. Available: https://arxiv.org/abs/1706.03762

[14] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.

[15] A. Gu and T. Dao, "Mamba: Linear-time sequence modeling with selective state spaces," *arXiv preprint arXiv:2312.00752*, 2023. [Online]. Available: https://arxiv.org/abs/2312.00752

[16] A. Nerode, "Linear automaton transformations," *Proceedings of the American Mathematical Society*, vol. 9, no. 4, pp. 541–544, 1958.

[17] M. Cranmer, "Interpretable machine learning for science with PySR and SymbolicRegression.jl," *arXiv preprint arXiv:2305.01582*, 2023. [Online]. Available: https://arxiv.org/abs/2305.01582

[18] M. G. Kendall, "A new measure of rank correlation," *Biometrika*, vol. 30, no. 1/2, pp. 81–93, 1938.

[19] M. G. Kendall and B. B. Smith, "The problem of $m$ rankings," *The Annals of Mathematical Statistics*, vol. 10, no. 3, pp. 275–287, 1939.

[20] M. G. Kendall, "Partial rank correlation," *Biometrika*, vol. 32, no. 3/4, pp. 277–283, 1942.

# A. Team Roles

| Member | Responsibility |
|---|---|
| Siddardha Reddy | coordination; D4 representation probing, reproduction gate and causal check; rules engine, data and model interface; preregistration and amendments; scoring launcher and H1 to H3 analysis |
| Nikhil Obuleni | zoo trainer and throughput; D3 inductive bias probe; fragility suite; continuous integration; Streamlit demo |
| Viharika Vemparala | D1 next token validity; report skeleton; data and splits write up |
| Venkatesh Nagarjuna | D2 state equivalence review and sign off; proposal and progress reports; presentation |

Table 15: Main responsibilities. Each member trained and scored one slice of the zoo on their own instance; the analysis and the paper are shared.

# B. Reproducibility

Code, data references, results and tags are in the repository `https://github.com/nik-hill-323/fall-2026-group6`. From the repository root:

- train one slice of the zoo (slice number 0 to 4):
  `bash src/shellscripts/train_slice.sh`
- score one slice with every diagnostic and both fragility suites:
  `bash src/shellscripts/score_slice.sh`
- run the preregistered analysis (master table, result files and figures):
  `python src/analysis/hypotheses.py`
- run the exploratory analysis:
  `python src/analysis/exploratory_othello.py`
- run the exploratory whole game measures (on the instance that holds the checkpoints), then summarize them:
  `python src/analysis/sequence_othello.py -all -anchors`
  `python src/analysis/sequence_othello.py -summarize`

- make the figures of this report:
  `python reports/Latex_report/make_figures.py`

A plain language summary of every Othello result is in `results/analysis/othello/RESULTS.md`.